

SmartPet Collar

Technical Specification Document

v1.0

Product: SmartPet Advanced Tracking & Communication Collar

Classification: Wearable Pet Device with Real-Time Tracking

Target Market: Companion animals (dogs 15+ lbs)

Development Status: Engineering Specification Phase

Date: July 10, 2026

EXECUTIVE SUMMARY

The SmartPet Collar combines premium vegetable-tanned leather construction with integrated electronics for GPS tracking, audio communication, and behavioral notification. The system integrates a single unified circuit board that manages all functionality via Bluetooth LE connectivity to mobile platforms.

SECTION 1: MATERIAL SPECIFICATIONS

1.1 Outer Material

Property	Specification
Material	Vegetable-tanned leather (Aniline or semi-aniline grade)
Grade	Full-grain thickness 1.8-2.0mm
Weight Tolerance	±5% per finished piece
Color Options	Cognac, burgundy, black, tan (customizable)
Tensile Strength	≥25 MPa
Tear Resistance	≥3.5 N/mm (per ASTM D1938)
Durability Target	5+ years active use under normal conditions

1.2 Inner Liner Material

Material: Ultra-high molecular weight polyethylene (UHMWPE) or polytetrafluoroethylene (PTFE)
Thickness: 0.5mm ±0.1mm
Coefficient of Friction: ≤0.08 (dry), ≤0.15 (wet) vs. canine fur
Coverage: Full inner surface with thermally bonded adhesive (EVA-based, pet-safe)
Attachment Method: Water-based polyurethane adhesive (non-toxic, certified safe for animals)

1.3 Hardware

Buckle Type: Military-grade side-release buckle (plastic polymer, water-resistant)
Stitching: Waxed polyester thread, box-stitched with X-pattern reinforcement
Fastener Points: Stainless steel 316L rivets (4 per collar) at stress points
D-Ring: Stainless steel 304L (1.2" diameter, rated for 200 lbs pulling force)

SECTION 2: ELECTRONIC COMPONENTS & ARCHITECTURE

2.1 Core Microcontroller

Part: Nordic nRF52840 (ARM Cortex-M4, 64 MHz)
RAM: 256 KB
Flash Storage: 1 MB
Operating Voltage: 3.0-3.6V (two AA batteries)
Rationale: Excellent Bluetooth LE integration, low power consumption, proven in wearable market

2.2 GPS Module

Parameter	Specification
Part	ublox MAX-M8Q (or newer M9N)
Dimensions	11.8mm x 15.7mm x 3.5mm
Accuracy	±2.5m (90% of fixes, open sky)
Cold Start Time	25 seconds
Warm Start Time	1 second
Update Rate	10 Hz maximum (configurable)
Power Consumption	35 mA active, <10 µA standby
Frequency	L1/L5 dual-band

2.3 Audio System (Beeper/Speaker)

Speaker Type: Piezoelectric transducer (SMD speaker)
Dimensions: 11mm x 8.5mm x 3.8mm
Frequency Range: 2-5 kHz (optimized for canine hearing)
Sound Pressure Level: 85-92 dB @ 10cm
Current Draw: 25-40 mA when active
Tone Options: 5 distinct patterns (lost dog alert, training recall, emergency)

2.4 Vibration Motor

Part: Coin vibration motor (10mm diameter, 3mm height)
Type: ERM (eccentric rotating mass)
Voltage Rating: 3.0-3.3V
Vibration Intensity: 3000-4000 RPM adjustable via PWM
Force: 0.8-1.2 G max acceleration
Power: 30-50 mA when active
Pattern Flexibility: Pulse, continuous, rhythmic (customizable)

2.5 Bluetooth/Radio Module

Primary: Integrated in nRF52840 (Bluetooth 5.2 Low Energy)
Range: 200+ meters line-of-sight (50-100m through walls)
TX Power: +4 dBm (configurable to -30 dBm)
Data Rate: 2 Mbps (LE 2M PHY mode)
Advertising Interval: 100 ms (standard), configurable to 1000 ms

2.6 Audio Codec & Microphone (Calling Feature)

Microphone: MEMs microphone module (3.0 x 3.0 x 1.0 mm)

Sensitivity: -42 dBV/Pa

Frequency Response: 50 Hz - 15 kHz

ADC: 16-bit, 16 kHz sampling rate

Codec: OPUS codec (10-20 kbps bitrate)

Audio Quality: Sufficient for directional calling

SECTION 3: CIRCUIT BOARD ARCHITECTURE

3.1 PCB Specifications

Form Factor: Single unified board, compact rectangular (62mm x 38mm x 8mm total thickness)

Layer Count: 4-layer PCB (signal, power, ground, signal)

Material: FR-4 (standard epoxy-fiberglass)

Copper Weight: 1 oz internal, 2 oz external

Via Plating: 0.3mm minimum via diameter, gold ENIG finish

Mounting: PCB mounted inside laser-cut acetal copolymer (Delrin) chassis

Connector: USB-C for charging/firmware updates (5V/1A standard)

3.2 Power Distribution

Battery Terminals: Spring-loaded contacts for 2x AA battery holder

Primary Rail: 3.3V (nRF52840, GPS, radio, audio codec)

Secondary Rail: 5V (buzzer driver, optional features)

Current Budget (idle): <100 μ A

Current Budget (GPS active): 50-65 mA continuous

Current Budget (calling mode): 80-120 mA

SECTION 4: FIRMWARE & SOFTWARE ARCHITECTURE

4.1 Operating System

Base: Nordic nRF5 SDK v17.1.0 (FreeRTOS real-time OS)
Bluetooth Stack: SoftDevice S140 (official Nordic BLE stack)
Bootloader: Secure bootloader with firmware update capability over Bluetooth

4.2 Core Firmware Modules

Module A - Bluetooth Manager: GAP/GATT profile, pairing/bonding (8 devices max), connection timeout 30 sec

Module B - GPS Manager: UART communication, NMEA parsing, position validation, geofence monitoring

Module C - Audio Manager: Tone generation, vibration control, audio playback, microphone sampling, echo cancellation

Module D - Communication Stack: Remote command parsing (JSON), status reporting, error handling

Module E - Power Management: Dynamic frequency scaling, GPS duty cycling, radio power states, battery monitoring

4.3 Application Protocol

Command Set (Mobile App to Collar):

GPS_ENABLE/DISABLE | BEEP_PATTERN [0-4] | VIBRATE_PATTERN [0-100] | START_CALL |
SET_GEOFENCE | LOCATION_REQUEST | BATTERY_STATUS_REPORT

Status Reports (Collar to Mobile App):

GPS position (lat/lon/accuracy/timestamp) | Battery percentage | Signal strength (RSSI) | Temperature |
Firmware version

SECTION 5: POWER SPECIFICATIONS

5.1 Battery Selection

Parameter	Alkaline AA	NiMH Rechargeable
Chemistry	Duracell LR6	2000+ mAh
Voltage	3.0V nominal (2x AA)	3.0V nominal (2x AA)
Capacity	2400-3000 mAh	1600-2000 mAh
Standby Runtime	45-60 days	N/A
Active Tracking	7-10 days	4-6 days per charge
Continuous GPS+Call	3-5 hours	2-3 hours per charge

5.2 Charging System (Optional Upgrade)

Battery: Rechargeable version uses 2x AA NiMH only
Charging Circuit: TP5100 dual-channel charger IC
Charge Rate: 500 mA per cell (1A total)
Protection: Overvoltage, short-circuit, thermal cutoff
Dock: USB-C connector with docking station
Charge Indicator: Single LED (red=charging, green=complete)

5.3 Power Profile Analysis

Scenario: Urban Dog (10 location checks/day, occasional calling)
GPS fix every 10 min: $35 \text{ mA} \times 25 \text{ sec} = 14.6 \text{ mAh}$ per fix = 511 mAh/day
BLE idle (advertising): 0.12 mAh/day
Calling (10 min total/day at 100 mA): 16.7 mAh/day
Standby losses: 5 mAh/day
Total: ~570 mAh/day → 4-5 days on alkaline, 3-4 days on NiMH

SECTION 6: PHYSICAL DESIGN & ERGONOMICS

6.1 Form Factor

Collar Width: Adjustable 10"-24" neck circumference
Thickness: 15mm (leather + hardware)
Electronics Module Height: 8mm above leather surface
Weight: 75-85 grams total (including batteries)
Attachment: D-ring positioned 1" from buckle for leash attachment

6.2 Enclosure Design

Material: Medical-grade acetal (Delrin) or polycarbonate
Dimensions: 62mm L x 38mm W x 8mm H
Mounting: Ultra-low-profile adhesive (3M VHB 5952)
Water Resistance: IP54 rating (splash-proof, not submersible)
Thermal Management: Minimal (low power dissipation <500 mW max)

6.3 User Interface

Single Mechanical Button: Firmware reset (hold 5 seconds)
LED Indicators:

- Blue: Bluetooth connected
- Green: GPS lock acquired
- Red: Low battery warning
- Yellow: Charging (if rechargeable version)

Tactile Feedback: Vibration motor for haptic user feedback

SECTION 7: CONNECTIVITY & NETWORK ARCHITECTURE

7.1 Bluetooth LE Specification

Protocol: Bluetooth Low Energy 5.2
Frequency Band: 2.4 GHz ISM (2402-2480 MHz)
Modulation: GFSK (Gaussian Frequency Shift Keying)
Channel Hopping: 37-39 advertisement channels
MTU Size: 247 bytes
Connection Interval: 15-1000 ms (configurable)

7.2 Mobile App Architecture

Target Platforms: iOS 13.0+ and Android 8.0+

Core Features:

- Real-time GPS tracking map (Google Maps / Apple Maps)
- Geofence alerts with push notifications
- Two-way audio calling
- Beep/vibrate control (5 patterns)
- Battery status dashboard
- Historical location playback (7-day retention)
- Multi-dog support (5 collars per account)

7.3 Website Dashboard

Real-time map view (auto-refresh 10 seconds)
Historical timeline with heatmap
Custom geofence drawing
Device settings (GPS interval, battery mode)
Account management (multi-user, sharing)
Data Export: CSV/JSON (30-day rolling history)

SECTION 8: AUDIO SYSTEM DETAILED SPECIFICATIONS

8.1 Beeper Acoustic Profile

Alert Mode	Frequency	Pattern	SPL @ 10cm
Lost Dog Alert	1 kHz	2s on / 1s off (continuous)	88 dB
Training Recall	4.2 kHz	500ms burst + 250ms gap (5x)	88 dB
Emergency Alert	2.5+4.2 kHz	Warbling chirp (250ms)	92 dB
Incoming Call	2-5 kHz	Ascending sweep (1 sec repeat)	88 dB

8.2 Speaker/Microphone Calling Specifications

Microphone Path:
16-bit/16 kHz analog capture | High-pass filter (100 Hz) | AGC with -6 dB/s attack | Noise gating (-40 dBFS) | OPUS encoding (12 kbps) | BLE transmission with 100ms jitter buffer

Speaker Path:
OPUS decoding (10-20 kbps) | Gain control (-12 to +12 dB) | Low-pass filter (8 kHz) | Speaker driver (200-500 mW peak) | Audio latency target: <300ms one-way

8.3 Acoustic Testing Requirements

Frequency Response: ± 3 dB from 1-15 kHz (anechoic chamber)
THD (Total Harmonic Distortion): <5% at max volume
Crosstalk: Speaker-to-microphone isolation >40 dB
Robustness: Operation in 0-45°C ambient temperature range

SECTION 9: SAFETY & COMPLIANCE SPECIFICATIONS

9.1 Material Safety

Leather: OEKO-TEX Standard 100 compliance (no harmful chemicals)
Adhesives: Non-toxic polyurethane, pet-safe (EN 71-3 compliant)
Piezo Elements: Non-lead solder (RoHS 2.0 compliant)
Vibration Motor: Sealed housing (no pinch hazard)

9.2 Electrical Safety

Voltage: 3V maximum (below shock hazard threshold)
Current Limiting: All outputs <100 mA continuous
No Electrical Shock: Vibration and audio are purely sensory alerts
EMC Compliance: FCC Part 15 (Bluetooth emissions)

9.3 Functional Safety

Geofence Tolerance: ± 5 meters (accounts for GPS variance)
Lost Dog Mode: Continuous audible alert if geofence breached
Battery Low Protection: Mandatory app alert when <10%
Water Resistance: IP54 (splash-proof, not submersible)

9.4 Standards Compliance

FCC ID: Required registration for Bluetooth transceiver
CE Mark: European Conformity (European distribution)
IC Canada: Device licensing (Canadian market)
Pet Wearable Safety: Follow ASTM F2658 wearable electronics best practices

SECTION 10: PERFORMANCE & TESTING SPECIFICATIONS

10.1 GPS Accuracy Testing

Cold Start Accuracy: 95% within 5m (suburban/rural), 10m (urban)
Warm Start Time: <1 second
Acquisition Rate: 1 Hz (configurable to 0.2 Hz)
Multipath Mitigation: High-sensitivity L1/L5 dual-band
Urban Performance: SAT-Based DGPS correction (optional)

10.2 Bluetooth Range Testing

Open Air Range: 200+ meters line-of-sight
Obstacle Range: 50-75 meters (walls/trees)
Reliability: <1% packet loss at 50m line-of-sight
Interference Robustness: Operation maintained with 5+ other Bluetooth devices

10.3 Battery Runtime Validation

Test Condition	Expected Runtime (Alkaline)	Expected Runtime (NiMH)
Idle (BLE advertising)	45-60 days	N/A
Active Tracking (GPS every 5 min)	7-10 days	4-6 days
Continuous Calling	3-5 hours	2-3 hours

10.4 Audio Quality Metrics

Microphone SNR: >40 dB
Speaker Frequency Response: ±6 dB (500 Hz - 8 kHz)
Latency: <500 ms round-trip
Codec Performance: OPUS at 12 kbps = 160 ms block, ~20 ms algorithmic delay

SECTION 11: MANUFACTURING & ASSEMBLY

11.1 Bill of Materials (BOM)

Component	Unit Cost	Notes
nRF52840 microcontroller	\$6-8	Primary processor
ublox MAX-M8Q GPS module	\$18-22	GPS receiver
Ceramic GPS antenna	\$2-3	L1/L5 dual-band
Piezoelectric speaker	\$1-2	Audible alerts
Vibration motor (10mm coin)	\$2-3	Tactile feedback
MEMs microphone module	\$1-1.50	Voice capture
Bluetooth antenna components	\$0.50	PCB integrated
Power management circuit	\$2-3	Battery management
USB-C connector	\$0.50	Charging/firmware
PCB (4-layer FR-4)	\$8-12	Electronics substrate
Acetal enclosure (CNC)	\$5-7	Component housing
Vegetable-tanned leather	\$15-20	Full hide material
PTFE liner	\$2-3	Friction reduction
Buckle and hardware	\$3-4	Attachment points
Stitching and rivets	\$1-2	Assembly materials
AA battery holder	\$1-1.50	Power storage
Spring contacts/wiring	\$2-3	Electrical connectivity
Assembly labor (semi-auto)	\$8-12	Per unit labor cost

Total BOM Cost: \$78-106 per unit (1,000 unit volume)

Target Retail Price: \$199-299 (2-3x markup)

11.2 Assembly Process

1. PCB population (pick-and-place machine, reflow oven)
2. Firmware flashing and testing (functional test rack)
3. Enclosure assembly (adhesive bonding, press fit)
4. Leather cutting and stitching (hand-sewn or semi-automated)
5. Electronics integration into leather (adhesive mounting)
6. Hardware installation (riveting D-ring)
7. Quality assurance (electrical test, GPS lock verification, BLE pairing)
8. Packaging and calibration (app setup guide included)

SECTION 12: FUTURE ENHANCEMENTS & ROADMAP

Phase 1 (Current): Core Tracking + Calling

- GPS tracking with geofence alerts
- Two-way audio calling
- Vibration/beep alerts (5 patterns)
- Mobile app + web dashboard

Phase 2 (Q2 2025): Advanced Features

- Health metrics (heart rate via PPG sensor)
- Activity level tracking
- Barking detection (audio classification)
- Social network for dog owners
- Rechargeable battery option with USB-C dock

Phase 3 (Q4 2025): AI Integration

- Escape prediction algorithm (behavior analysis)
- Smart recall training (rewards-based audio cues)
- Terrain analysis (water/brush alerts)
- Behavior anomaly detection

Phase 4 (2026+): Advanced Connectivity

- LTE/5G fallback for extended range
- Mesh networking (collar-to-collar relaying)
- Environmental sensing (air quality, UV, temperature)
- Smart home integration (door locks, alerts)

END OF TECHNICAL SPECIFICATION

Document Version	v1.0
Last Updated	July 10, 2026
Prepared by	Engineering Division
Classification	Technical Reference Document
Status	Engineering Specification Phase